

Day 09 Hybrid Notes

Similar Triangles in Diagrams and Contexts | Cloze + Outline

Name:	Date:	Period:
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Learning Targets
I can identify corresponding sides in similar triangles.
I can set up a correct proportion from a diagram or context.
I can use scale factor to solve for missing side lengths.

Part 1: Cloze Notes - Essential Ideas
1. Similar figures have the same _____ but not always the same _____.
2. Corresponding sides are sides in the same _____ in two similar figures.
3. Scale factor = _____ length / _____ length.
4. If two triangles are similar, corresponding side lengths are _____.
5. A proportion says two _____ are equal.
6. AA Similarity means two triangles are similar if two pairs of _____ match.
7. Before solving, mark matching _____ and corresponding _____.

Quick Reference Routine 1. Mark the matching parts. 2. Choose matching sides. 3. Write ratios in the same order. 4. Solve the proportion. 5. Check if the answer makes sense.	Part 2: Outline Notes - Big Picture I. Similar means same shape. A. Matching angles are _____. B. Matching sides are _____. II. This connects to prior math: A. Algebra: proportions and cross products. B. Coordinates: dilations and scale factor. C. Real life: maps, shadows, photos, models.
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The Hardest Part
Usually the hardest part is not the arithmetic. The hardest part is figuring out which sides _____ and keeping ratios in the same _____.

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Guided Setup + Active Learning

Part 3: Guided Setup Practice

Triangle ABC is similar to triangle DEF. $AB = 8$, $DE = 20$, $BC = 10$, $EF = x$.

1. I am trying to find: _____
2. Match the sides: $AB \leftrightarrow$ _____ and $BC \leftrightarrow$ _____
3. Scale factor from ABC to DEF: $20 / 8 =$ _____
4. Proportion: $8 / 20 = 10 / x$
5. Cross multiply: $8x =$ _____
6. Solve: $x =$ _____
7. Sentence answer: The missing side length is _____.

Part 4: Error Analysis

A student writes: $8 / 10 = 20 / x$

What went wrong?

Correct proportion:

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Before You Solve Checklist

- ☐ I marked matching angles.
- ☐ I matched corresponding sides.
- ☐ I wrote ratios in the same order.
- ☐ I checked whether my answer makes sense.

Part 5A: Draw It

Draw two similar triangles. Label two matching angles, three side lengths, one unknown side, and the scale factor.

Part 5B: Two-Sentence Summary

Sentence 1: Similar triangles help us because _____.

Sentence 2: To solve a missing side problem, I need to _____.

Part 5C: Application Problem

A 5-foot student casts a 7-foot shadow. A tree casts a 35-foot shadow.

Proportion: _____

Tree height: _____

Sentence answer: _____